

Databricks File System (DBFS)

DBFS is the core and the heart of databricks, because this is the way through which we're going to read and write the data.

What is DBFS?

DBFS is a distributed file system that abstracts the storage infrastructure in databricks. It provides a unified interface for accessing and managing the data in the databricks workspace. DBFS helps us users to interact with the files which are available in the databricks. Basically, it works like a local file system to the databricks.

DBFS is actually built on top of a cloud (AWS, Azure). So under the hood, your data might get stored either on Azure Data Lake Storage, AWS, or a Google Cloud Storage based on the cloud you have chosen.

The below are some of the key features of DBFS:

1. Unified interface
2. Seamless integration
3. Access to cloud storage
4. Scalable and distributed
5. Data persistence
6. Multiple Storage

DBFS Path Structure

DBFS uses a Unix-like directory structure. The parts in the DBFS can be either:

- Absolute paths: begin with /
 - Example: /mnt/data/sample.csv
- Relative paths: relative to the current working directory
- DBFS prefix: access programmatically using the dbfs:/ prefix
 - Example: dbfs:/mnt/data/sample.csv

DBFS Usecases

Some of the important use cases of DBFS are:

1. Data loading
2. Data writing
3. Backup and archiving
4. Collaboration

DBFS Limitations

Some of the important limitations of DBFS are:

1. Dependence on cloud storage
2. Path restrictions